

2. Data

2.1 Source and Description

The data are sourced from the Øresund Bridge operator's publicly available monthly traffic statistics (SOURCE). The variable of interest is the number of passenger car crossings per month, recorded from July 2000 through March 2026, yielding $T = 309$ observations with no missing values. The unit is an absolute vehicle count per month, i.e. crossings to and from Sweden/Denmark.

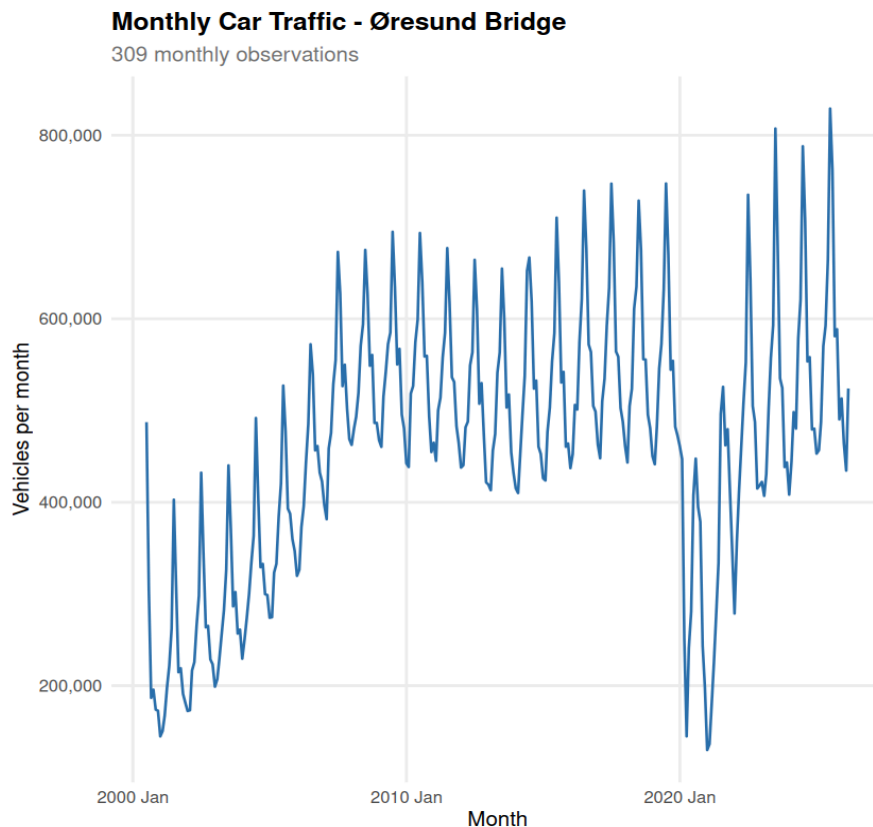


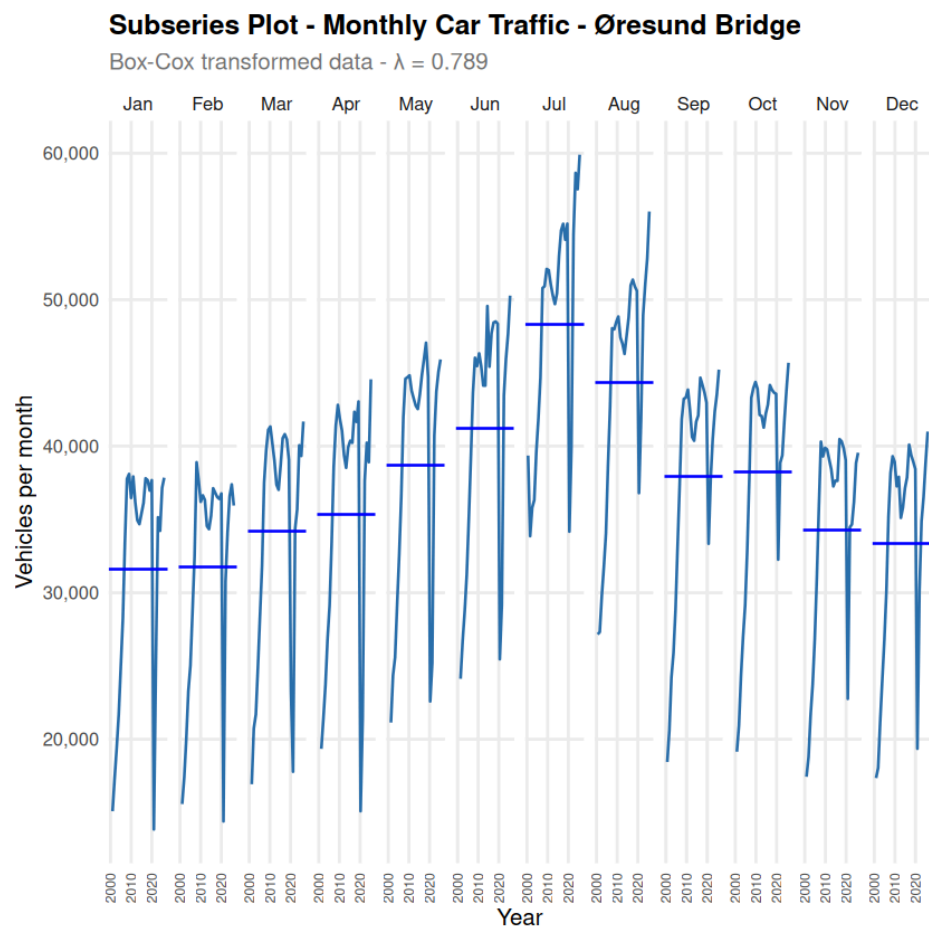
Figure X presents a plot of the raw Cars series. Three structural features are immediately apparent. First, a strong upward trend which levels out after ~ 2007 , with monthly counts rising over the two-decade period. Second, a strong and increasing annual seasonal cycle is superimposed on the trend, producing sharp peaks in the summer months and troughs in winter each year. Third, a dramatic collapse in traffic is visible from early 2020, the fingerprint of COVID-19 border restrictions that temporarily severed cross-border mobility between Sweden and Denmark.

2.2 Main Characteristics

Three structural features dominate the series and drive every subsequent modelling decision.

Long upward trend

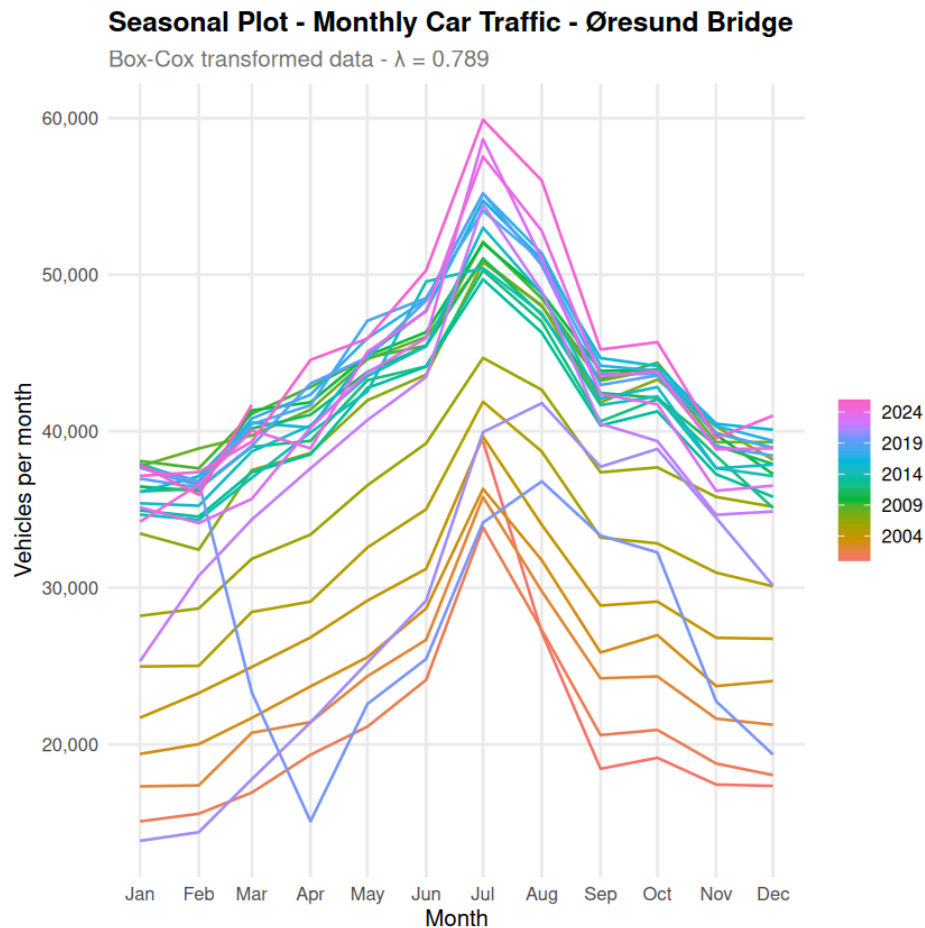
Traffic peaks have grown for roughly two decades following the bridge's opening. The subseries plot of the Box-Cox transformed series (Figure X) reveals that this growth was not uniform across months: summer peaks show a sustained increase throughout the period, while winter months have remained roughly steady or even decreased slightly. Notable for all months is a sharp increase in the first years, correlating to the strong initial trend, and clear dip during the Covid-19 pandemic.



Strong annual seasonal cycle

The seasonal plot of the Box-Cox transformed series (Figure X) shows an inverted-U shape, peaking in July each year and falling to a winter trough in January–February, consistently across all 25 years (with the exception for the Covid-19 pandemic). The curves for each year overlay closely in shape — the same inverted-U pattern, peaking in July, appears in every year of the sample. The amplitude is not constant, however: summer months grow faster than winter months, and this differential growth persists even in the Box-Cox

transformed series. The upward shift of the lines over time reflects the long-run growth in traffic, with the progression visibly flattening from around 2007.



COVID-19 disruption

An abrupt collapse in traffic is visible from March 2020, with the series recovering only gradually through 2021 and returning to its pre-pandemic trajectory by mid-2022. April 2020 recorded 144,759 car crossings — the single lowest monthly count in the entire sample — representing approximately a 70% drop relative to seasonal expectation. The identification and treatment of this disruption are detailed in Section 2.6.

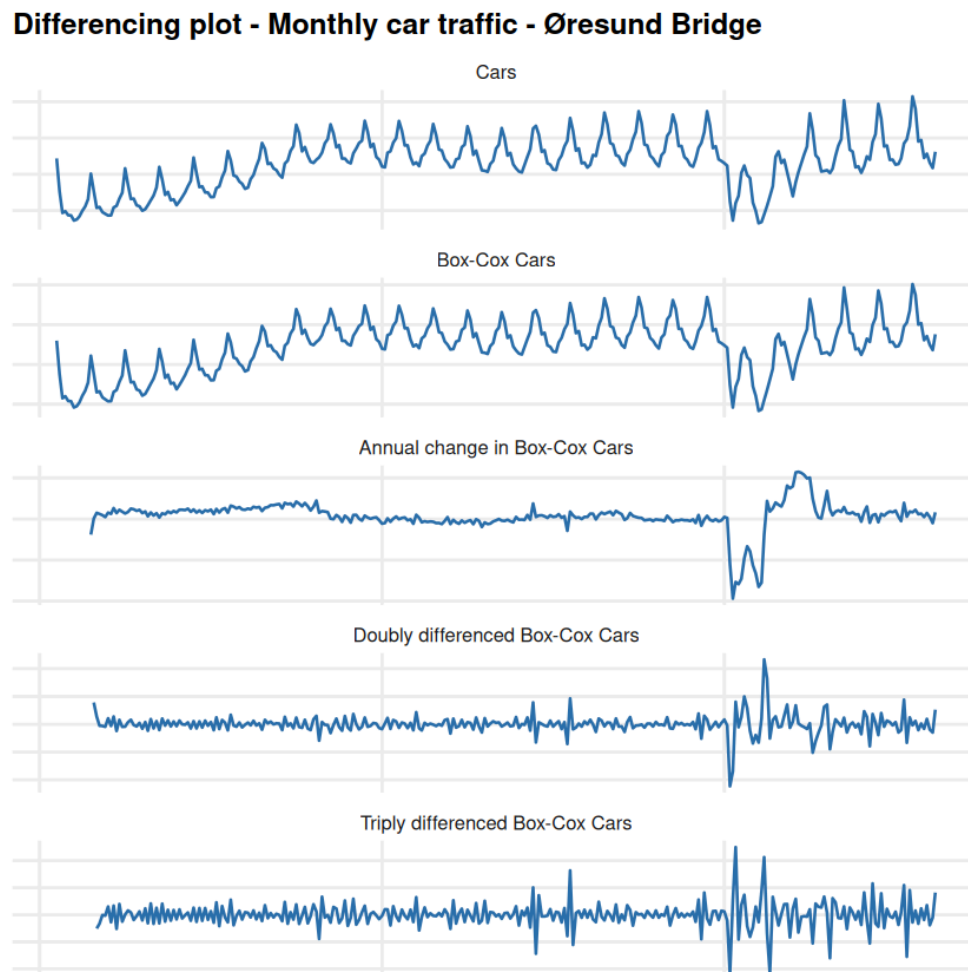
2.3 Variance Stabilisation and Transformation

The raw series exhibits heteroscedastic seasonal fluctuations: summer peaks grow in absolute magnitude as the overall level of traffic rises, while winter troughs remain comparatively flat. This variance-level dependence violates the assumption of homoscedastic errors required for valid prediction intervals and motivates a functional transformation before modelling.

Guerrero's method, applied to the Cars series, selects $\lambda = 0.7899$. This value is meaningfully different both from $\lambda = 1$, which would correspond to no transformation, and from $\lambda = 0$, which would correspond to a log transformation. Box-Cox($\lambda = 0.79$) compresses the large

summer values relative to winter troughs. The transformation reduces this differential, as visible in the transformed seasonal plot, but does not eliminate it.

Figure X presents a five-panel differencing plot illustrating the effect of successive transformations.



(IMAGE: Multi-panel differencing plot — Cars (scaled), Box-Cox Cars, Annual difference $D=1$, Doubly-differenced $d=1$ $D=1$, Triply-differenced $d=2$ $D=1$)

The top panel shows the raw Cars series, exhibiting both the non-constant variance and the upward trend. The second panel shows the Box-Cox transformed series, where the seasonal amplitude is reduced relative to the raw series. The third panel applies a seasonal difference at lag 12, which removes the annual cycle and renders the long-run trend approximately stationary, establishing $D = 1$. The fourth panel adds a first-order regular difference, removing the remaining trend and yielding a series that oscillates around zero with no discernible drift, establishing $d = 1$. The fifth panel applies a second regular difference ($d = 2$, $D = 1$): the result is more volatile than the fourth panel, confirming that one regular difference is sufficient and that over-differencing would introduce unnecessary noise. This progression simultaneously motivates $d = 1$ and $D = 1$.

2.4 Stationarity

Stationarity is assessed on the raw Box-Cox transformed series using two complementary unit-root tests with opposite null hypotheses. The KPSS test has the null of stationarity; the ADF test has the null of a unit root.

Test	Statistic	p-value	Conclusion
KPSS (H_0 : stationary)	1.6096	0.01	Reject → non-stationary
ADF (H_0 : unit root)	-2.9251	0.186	Fail to reject → non-stationary

The two tests agree: the raw Box-Cox transformed series is non-stationary. KPSS rejects the null of stationarity ($p \leq 0.01$), driven by the upward trend. ADF fails to reject the unit-root null ($p = 0.186$). The automated selectors `unitroot_ndiffs = 1` and `unitroot_nsdiffs = 1` confirm that $d = 1$ and $D = 1$ are required.

After applying $d = 1$ and $D = 1$ (the doubly-differenced Box-Cox series), the same tests yield:

Test	Statistic	p-value	Conclusion
KPSS	0.02957	0.10	Fail to reject → stationary
ADF	-6.9127	0.01	Reject unit root → stationary

Both tests now agree: the doubly-differenced Box-Cox series is stationary. The trend is treated as stochastic — that is, as a unit root — consistent with standard practice for economic time series of this length. A stochastic trend allows the level to drift over the forecast horizon, which better captures long-range forecast uncertainty than a fixed deterministic trend would.

2.5 Seasonality

The seasonal plot and subseries plot of the Box-Cox transformed series (Figures X and X) confirm a strong 12-month cycle across all 25 years. The seasonal shape is consistent — the same inverted-U profile, peaking in July, appears in every year of the sample. The seasonal amplitude, however, drifts stochastically: as visible in the transformed seasonal plot, the summer-to-winter gap widens gradually over the sample. This is confirmed formally by `unitroot_nsdiffs = 1`, indicating a seasonal unit root, and by the dominant spike at lag 12 in the ACF of the Box-Cox transformed series. Seasonal differencing ($D = 1$) resolves this non-stationarity.

2.6 COVID-19: Structural Break Analysis and Dummy Design

Identifying the outlier period

A robust STL decomposition (`robust = TRUE`, `trend window = 13 months`) was applied to the raw Cars series. The robust option causes the STL algorithm to downweight extreme observations when fitting the trend and seasonal components, pushing the COVID-19 signal

cleanly into the remainder. Z-scores were then computed on the remainder (subtracting its mean and dividing by its standard deviation), and months with $|z| > 2$ were flagged as outliers.

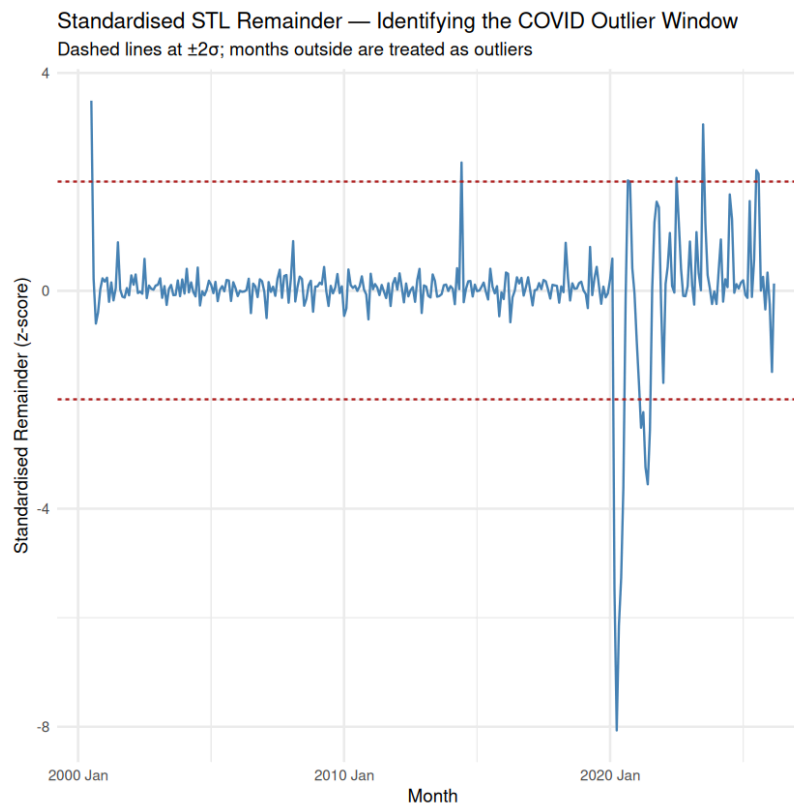


Figure X plots the standardised STL remainder over the full sample, with dashed lines at $\pm 2\sigma$. The most severely affected months, ranked by most negative z-score, are: April 2020 ($z = -8.07$, Cars = 144,759), May 2020 ($z = -6.14$, Cars = 241,158), March 2020 ($z = -5.47$, Cars = 251,316), and June 2020 ($z = -5.27$, Cars = 280,692). Together with the remaining months flagged below -2σ through July 2021, these define the contamination window. The period dummy is accordingly set to 1 for March 2020 through July 2021 inclusive — a window of 17 months. The plot also reveals several positive outliers in the post-COVID period, most prominently July 2023 ($z = +3.05$) and July 2025 ($z = +2.21$), reflecting genuine above-trend demand recovery rather than a data anomaly. These are not controlled for.

Is the COVID disruption a permanent structural break? A QLR test (Andrews, 1993, supremum-F) was applied to the doubly-differenced Box-Cox series, scanning all candidate breakpoints within the trimmed window (15% from each end of the sample). The F-statistic profile peaks within the COVID period — the intuitively correct location — but reaches only $\text{sup.F} = 2.369$, $p = 0.697$, far below the 5% critical value of approximately 8.68. Structural stability is therefore strongly confirmed at every candidate breakpoint date. As a supplementary cross-check, a single-point Chow test at August 2021 — the point at which the series had substantially recovered — returns $F = 0.059$, $p = 0.808$, fully consistent with

the QLR result. COVID-19 was a temporary, externally-driven shock that did not alter the underlying data-generating process. The full 25-year history is accordingly used for estimation.

Dummy variable design for ARIMA

Four dummy variables are constructed. The period dummy covid equals 1 for the entire March 2020 – July 2021 window (17 months) and 0 otherwise. Three impulse dummies — `imp_2020_apr`, `imp_2020_may`, and `imp_2020_mar` — each take the value 1 for their respective single month and 0 otherwise, corresponding to the three most extreme months by z-score: April 2020 ($z = -8.07$), May 2020 ($z = -6.14$), and March 2020 ($z = -5.47$). The logic of the design is as follows. The period dummy absorbs the average disruption across all 17 COVID months. The three impulse dummies capture the additional severity of the most extreme lockdown months beyond that average. Months within the period but outside the impulse list receive only β_{covid} ; March, April, and May 2020 each receive β_{covid} plus the respective impulse coefficient. For the out-of-sample forecast, all dummy values are set to zero. We later test our best performing ARIMA model with:

- i: All dummy variables
- ii: No dummy variables
- iii: Only period covid dummy

ETS cannot accommodate external regressors, and this incompatibility is one of the reasons ARIMA is treated as the primary model.

3. Methodology

3.1 Modelling Strategy

Two complementary model classes are applied. ARIMA explicitly models the autocorrelation structure of the doubly-differenced Box-Cox series and accommodates external COVID regressors — a capability ETS does not possess. ETS handles trend and seasonality through adaptive exponential smoothing without pre-differencing, as the methodological contrast.

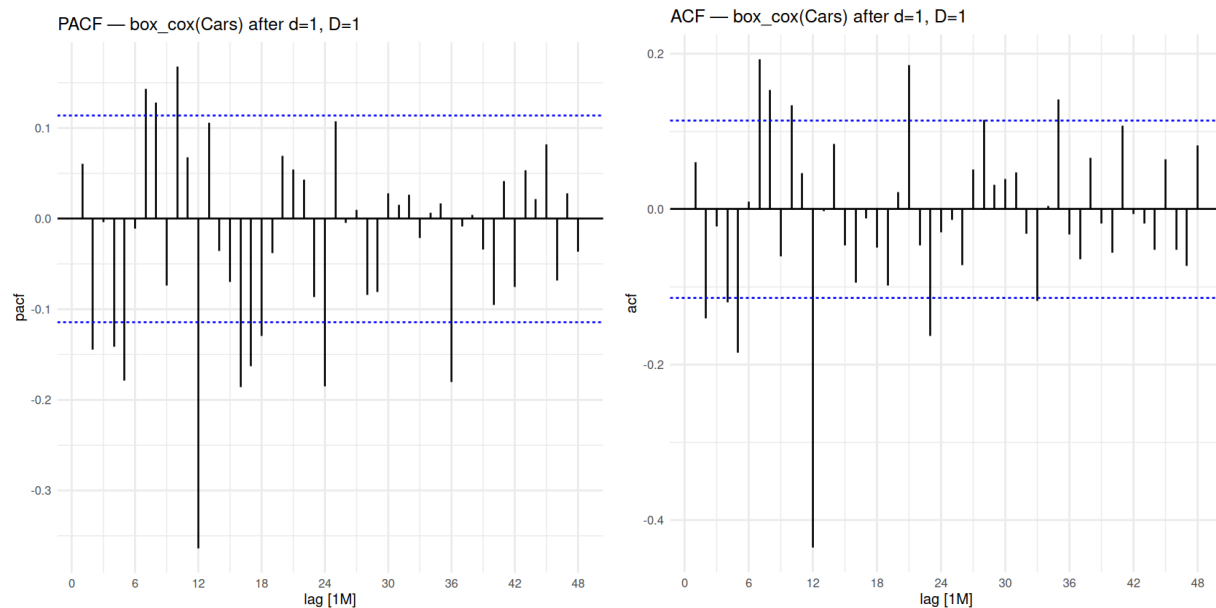
Within each class, AIC guides initial candidate comparison; expanding-window cross-validated RMSE at $h = 6, 12$, and 24 months is the primary selection criterion. AIC cannot compare across classes due to different likelihood normalizations. Both models are trained on July 2000 – March 2024, evaluated on a held-out test window of April 2024 – March 2026, and refit on the full dataset before producing the final forecast.

3.2 Model 1: SARIMA with COVID Regressors

Order identification.

Three candidates are estimated alongside an unconstrained automatic search: ARIMA(0,1,0)(0,1,1)[12] (seasonal ACF/PACF baseline), ARIMA(2,1,2)(0,1,1)[12] (manual identification), ARIMA(2,1,3)(0,1,1)[12] (extended MA candidate), and `auto.arima` with `stepwise = FALSE`, `approximation = FALSE`.

Seasonal part — PDQ = (0,1,1). The ACF of the doubly-differenced Box-Cox series shows a single large negative spike at lag 12 decaying smoothly at lags 24 and 36 — the textbook signature of a seasonal MA(1) process after one seasonal difference, consistent with the $D = 1$ requirement established in Section 2.



Non-seasonal part — $pdq = (2,1,2)$. After controlling for seasonal structure, the non-seasonal ACF and PACF exhibit an oscillating, geometrically decaying pattern across several lags — the signature of a mixed ARMA process, not a clean AR or MA cutoff. AR(2) and MA(2) terms provide the parsimonious description. Crucially, `auto.arima` independently selects the identical specification, so manual ACF/PACF identification and information-criterion search converge by two separate routes — strong validation of the order choice. ARIMA(2,1,3) is tested on cross-validation grounds (Section 4.3).

COVID dummy design. Without explicit control, 17 months of severe disruption would potentially distort all ARMA coefficient estimates. Four regressors are passed via `xreg`:

- `covid`: a period dummy equal to 1 for March 2020 – July 2021, absorbing the average level shift.
- `imp_2020_apr`, `imp_2020_may`, `imp_2020_mar`: impulse dummies for the three most severely affected months (STL remainder z-scores of -8.07 , -6.14 , and -5.47), capturing the additional severity beyond the period average.

All dummy values are set to zero over the forecast horizon. The COVID disruption is confirmed to be temporary, not a structural break: the QLR test yields $\text{sup.F} = 2.369$ ($p = 0.697$), well below the 5% critical value of ~ 8.68 , and a supplementary Chow test at August 2021 returns $F = 0.059$ ($p = 0.808$). The full 25-year sample is used for estimation without truncation.

SHOULD WE ADD THAT COVID IS TESTED FOR HERE?

Key assumptions. COVID regressors are treated as exogenous (border closures were set by public health authorities independently of traffic). Structural stability is supported by the QLR and Chow results. The Box-Cox transformed, differenced series is assumed approximately normal — supported by the near-Gaussian residual histogram — as required for valid prediction intervals.

3.3 Model 2: ETS(A,Ad,A)

Candidates. Five specifications are estimated: three STL+ETS variants (`stl_auto`, `stl_AAN`, `stl_AAdN`) that apply robust STL decomposition (`robust = TRUE`, `trend(window = 21)`, `season(window = "periodic")`) before fitting ETS to the seasonally adjusted series, and two direct ETS models on the full Box-Cox series (`ets_auto`, `ets_AAdA`).

Why ETS(A,Ad,A) over the auto selection. Applied directly to the Box-Cox series, the automatic search selects ETS(A,N,A) — no trend. This is overridden: the series has a clear 25-year upward trajectory and a no-trend model produces flat forecasts inconsistent with the data. The raw series shows a strong upward trend that decelerates markedly from around 2007, with growth broadly flattening in the decade before Covid-19; decelerating rather than linear growth, which motivated a dampened additive trend (Ad). The estimated damping parameter $\phi = 0.98$ confirms slow but non-negligible attenuation over the forecast horizon.

Error and seasonal components. Additive errors (A) are appropriate because Box-Cox transformation has already reduced the seasonal variance differential, as established in section 2.3. Additive seasonality (A) follows from the reduced seasonal amplitude in the transformed series, the remaining stochastic amplitude drift is addressed by $D = 1$ in ARIMA, while in ETS the near-zero estimated seasonal smoothing parameter $\gamma \approx 0.0001$ confirms that seasonal indices are effectively fixed across years.

The $\alpha \approx 1$ result. The estimated level parameter $\alpha = 0.9999$ means the level resets almost entirely to each new observation. A practical consequence: COVID observations from 2020–2021 are exponentially discounted to negligible weight by 2026 regardless of STL pre-processing. This explains why all three STL variants offer no accuracy advantage over plain ETS(A,Ad,A) — exponential discounting already achieves the same effective outlier downweighting (see Section 4.4).

Why ETS is the comparison model. All five ETS variants fail the Ljung-Box test (all $p < 0.001$). ETS has no mechanism to absorb residual autocorrelation — structurally the role of AR and MA terms. The resulting prediction intervals, which assume uncorrelated errors, are statistically invalid and likely too narrow. ETS is retained as the required second method, with ARIMA as the primary recommendation.

3.4 Forecast Evaluation Strategy

Cross-validation. An expanding-window CV is applied using `stretch_tsibble` with `.init = 120` months (minimum ten years of training data) and `.step = 12` (one fold per year, ~15 folds total). The same folds are used for both ARIMA and ETS to ensure a fair comparison. Accuracy is computed at $h = 6, 12$, and 24 months.

Metrics. Cross-validated RMSE is the primary selection criterion. The held-out test window additionally reports MAE, MAPE, and MASE. MASE scales MAE against a seasonal naïve benchmark — values below 1 indicate the model outperforms naïve seasonal forecasting.

Prediction intervals. 80% prediction intervals are reported throughout, following Hyndman & Athanasopoulos (2021) (SOURCE). At a 24-month horizon, 95% intervals span several hundred thousand vehicles and are practically uninformative; 80% intervals balance interpretability with uncertainty quantification. Interval validity rests on the white-noise residual assumption, assessed via the Ljung-Box test for each fitted model. Where this assumption is violated, prediction intervals are treated as indicative only rather than statistically valid.

